W11

Home ranges and animal movement

R PROGRAMMING STATISTICS

GPS collars record where an animal is at set intervals. From those fixes we estimate home ranges, describe movement and study which habitats it uses relative to those available. We work with three simulated common genets (Genetta genetta).

DURATION	About 2 hours.
PACKAGES	`adehabitatHR` (with `adehabitatLT`), `sf`, `terra`, `tidyverse`
FILES	`week11_home_ranges.R`, `week11_solutions.R`, `genet_gps.csv`, `genet_animals.csv`, `landcover.tif`, `landcover_classes.csv`, `montfred.gpkg`
DATA	1,042 fixes from 3 genets, one every 2 h for 30 days in March 2026, with some missed fixes.

Learning outcomes

- check telemetry data: fix intervals and gaps;
- estimate home ranges with the minimum convex polygon (MCP) and kernel density (KDE);
- calculate step lengths, turning angles and activity rhythms;
- compare habitat use and availability with selection ratios.

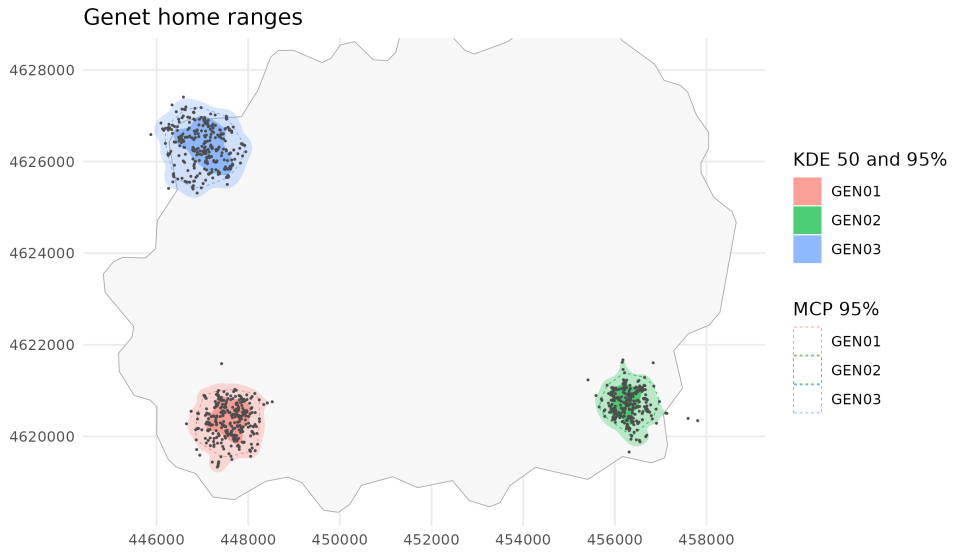
1 · Check the data

First: how often is there a fix, and how many are missing? Gaps (GPS failures under dense cover, for example) are not random and can bias analyses. Always plot the tracks before calculating anything.

2 · Home ranges

The **MCP** joins the outermost fixes; the 95% MCP drops the 5% furthest from the centre. It is simple and comparable across studies but includes areas the animal never uses. **KDE** estimates a utilisation surface: the 95% contour is the home range and the 50% contour the **core area**. Its result depends heavily on the bandwidth (h).

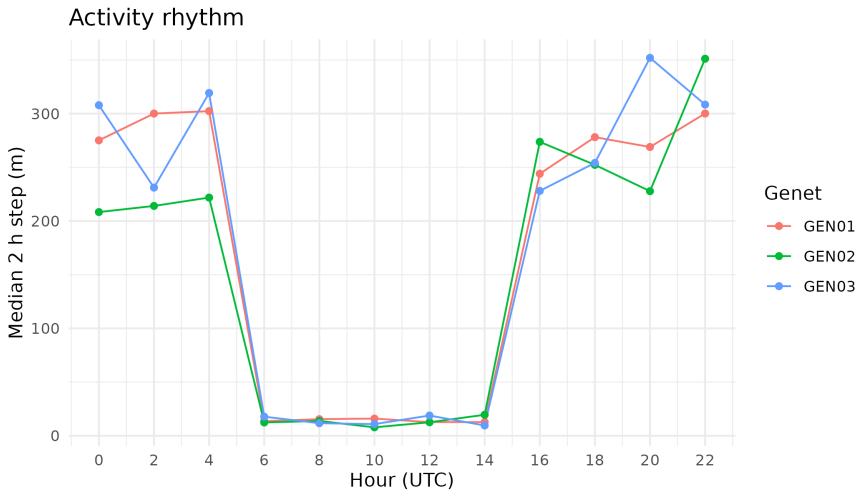
Genet	Sex	MCP 95% (ha)	KDE 95% (ha)	KDE 50% (ha)
GEN01	female	172	245	69
GEN02	male	120	170	44
GEN03	male	227	334	102



50% KDE (solid fill), 95% KDE (light fill) and 95% MCP (dashed line) of the three genets.

3 · Movement

`as.ltraj()` turns the fixes into a trajectory: for each step it gives the distance (`dist`), the interval (`dt`) and the turning angle (`rel.angle`). The median 2 h step is about 18 m by day and about 275 m at night: genets are nocturnal.



Median 2 h step length by hour of day (UTC).

4 · Habitat selection

We compare the proportion of fixes in each habitat (use) with that habitat's proportion inside the animal's 100% MCP (availability). A use/availability ratio > 1 means selection and < 1 avoidance (Johnson's third-order selection, 1980). With these simulated data the ratios are close to 1: the animals use forests roughly in proportion to availability. With real data, analyse the ratios with confidence intervals or resource selection functions (RSFs).

Scale and availability

The conclusion depends on what counts as available: the study area (second order), the home range (third order) or the surroundings of each step (step selection, SSF). Always state which scale you use.

Exercises

Answers in `week11_solutions.R`.

1. Recalculate the KDEs with `h = "LSCV"`. Larger or smaller areas?
2. How much do the 95% KDEs of GEN02 and GEN03 overlap?
3. What total distance did each genet travel in the month?
4. 95% MCP with night and day fixes separately.
5. Which habitat does each animal select most?

Before Week 12

- Install `install.packages("landscapemetrics")`.
- Prepare your final project (3–5 slides): see the Week 12 handout.